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Bachelor Thesis in Physics
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20XX

About ...

This Bachelor Thesis has been carried out by XYZ at the
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Nomenclature

Physical nomenclature:

- SKA: Square Kilometer Array
- EoR: Epoch of Reionization
- SM: Standard Model
- WDM: Warm Dark Matter
- ΛCDM: Lambda Cold Dark Matter
- IGM: Intergalactic Medium
- FRW: Friedmann-Robertson-Walker

Mathematical/statistical nomenclature:

- SBI: Simulation-Based Inference
- CNN: Convolutional Neural Network
- MLP: Multilayer Perceptron
- KL: Kullback–Leibler divergence
- CLT: Central Limit Theorem
- MI: Mutual Information
- CE: Conditional Entropy

Notation

Standard probability theory notation is adopted. In particular, the following notations are used:

- X for random variables with sample space $\mathcal{X}$
- $\mathcal{P}(\mathcal{X})$ for the set of all probability measures on the Borel σ-algebra over $\mathcal{X}$
- $P \in \mathcal{P}(\mathcal{X})$ for probability measures
- p_X for the density function of a probability measure P, index might be dropped if notation is clear
- $X \sim p$ for a random variable X whose distribution is described by p

When discussing neural networks, the symbol q or q_θ will be used for the model of a true distribution p.

1 Physics Background

I am currently still fighting with the citations, that will hopefully soon be resolved.

The current cosmological standard model predicts the epoch of Cosmic Dawn (CD) and the subsequent Epoch of Reionization (EoR) as mechanisms for star and structure formation and the following ionization of the intergalactic medium (IGM) at a redshift range of roughly $z \sim [5-20]$. A number of sources from multi-messenger astronomer have contributed to this current model: Scattering of CMB photons on free electrons during and after reionization can be detected once IGM became ionized significantly enough. This places the beginning of EoR at $z \sim 8$, though with a large uncertainty Lymann-alpha forests in the spectra of distant quasars and galaxies are used to estimate the neutral hydrogen fraction at low redshifts, placing the end of the EoR at around add here citation. The JWST with an observational range of $z \sim$ ... measures earliest galaxies and will be able to additionally constrain the speed of galaxy evolution.

A comparatively new probe is that of the 21 cm hyperfine transition line of neutral hydrogen, which constitutes a background spectrum similar to that of the CMB. It is the only known probe with which the Dark Ages could be studied and also the one to study CD to the highest redshifts. First experiments have measured reionization profiles in terms of power spectra and signals of possible dark matter presence cite. Much better results are expected from the Square Kilometer Array (SKA) Observatory, which is a radio telescope currently in construction which will image the 21 cm line of the IGM at a redshift range of $z \sim [6-30]$ with unprecedented accuracy. Hopes are that this signal can be used to significantly improve the current understanding of the reionization history and structure formation and possibly act as a new dark matter probe.

1.1 Theoretical Background

Still need some more subsections to explain the other parameters being examined

The Cosmological Setting

For the purposes of this thesis, a flat ΛCDM cosmology with an additional warm dark matter parameter is assumed. The ΛCDM-model is based on the Friedmann-Robertson-Walker metric:

$$ds^2 = -dt^2 + a^2(t)\left(\frac{dr^2}{1-kr^2} + r^2\big(d\theta^2 + \sin^2\theta d\phi^2\big)\right). \tag{1}$$

where $a(t)$ is the dimensionless scale factor and k is the curvature constant.

Combined with Einstein's field equations and the assumption that the universe may be described as a perfect fluid, the dynamics of this universe are described by the Friedmann equations:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3} - \frac{k}{a^2}, \tag{2}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p) + \frac{\Lambda}{3}. \tag{3}$$

where G is Newton's constant, Λ is the cosmological constant and ρ is the energy density of the universe. The scale factor as well as the energy density are time-dependent quantities.

Given the equation of state, $p = \omega\rho$, one finds for the density of different matter species:

$$\rho \sim \frac{\rho_0}{a^{3(1+\omega)}} \tag{4}$$

which yields the behavior of densities of

1. Radiation: $\rho_r \sim 1/a^4$
2. Matter: $\rho_m \sim 1/a^3$
3. Vacuum energy: $\rho_\Lambda \sim \text{const}$

Allowing matter, radiation and the cosmological constant to all be present, the Friedmann equation becomes

$$\left(\frac{\dot{a}}{a}\right)^2 + \frac{\kappa}{a^2} = \frac{8\pi G}{3}(\rho_m + \rho_r) + \frac{\Lambda}{3}. \tag{5}$$

By defining the Hubble parameter

$$H = \frac{\dot{a}}{a} \tag{6}$$

and dividing equation (5) by H^2, one can identify the three terms on the r.h.s with energy densities:

$$1 + \frac{\kappa}{H^2a^2} = \Omega_m + \Omega_r + \Omega_\Lambda \tag{7}$$

Knowledge of these density parameters therefore allows to determine the evolution of the scale factor a. Refer to conservation laws etc that allow to fix the densities

The ratio $H = \dot{a}/a$ is called the Hubble parameter and measures how fast the universe is expanding in the following sense: given two observers at $(t_1, r_1), (t_2, r_2)$ in a FRW-universe, where observer 1 emits a signal of duration Δt_1, observer 2 seeds a scale-factor dependent dilation of the signal duration, as measured by the cosmological redshift:

$$z = \frac{\Delta t_2}{\Delta t_2} - 1 = \frac{a(t_2) - a(t_1)}{a(t_1)} > 0 \quad \text{if universe expanding} \tag{8}$$

In the limit of $t_2 \to t_1$, if d is the instantaneous distance between the observers, one then has

$$z \approx \left.\frac{\dot{a}}{a}\right|_{t_1} d + \mathcal{O}((t_1 - t_2)^2) \tag{9}$$

such that H measures the expansion rate at a given time in units 1/s.

The Friedmann equations describe the global expansion history of the universe under strong symmetry assumptions. Their results match observations very well: citation They act as an effective theory though, since, for example, they contain no information about specific particles and their interactions with each other ausführen. Cosmology draws on other areas of physics to answer these questions. The following gives a short summary of the history of the universe when considering these as well.

Maybe just have a nice image of this long text below, it's taking up lots of space.

The first phase of the universe is marked by *radiation domination*. Immediately after the Big Bang, the universe underwent a period of rapid expansion called *inflation*, during which space expanded by appriximately an order of 10^{26}. This inflation was driven by the so-called inflaton field. During the following process of reheating, the energy of the inflaton field was converted to SM particles, which enabled subsequent *baryogenesis*, i.e. the creation of matter and antimatter particles. In the subsequent annihilation of matter.antimatter pairs, a small excess of matter paticles survived, composing the matter content of the universe today. There have been many proposed mechaninisms for both reheating and baryogensesis, and both are active fields of research. Insert sources here In the following time scale of $20 \cdot 10^{-12}\text{s} - 1000\text{s}$, particles acquired mass and coalesced into hadrons. Neurinos decoupled from baryons and thus the cosmic neutrino background was formed. What followed was an epoch named *big bang nuceosynthesis*, during which mostly protons and neutrons bound to primordial nuclei, consisting mostly of hydrogen, Helium-4 and traces of deuteron, Helium-3 and lithium.

The second phase of the universe is marked by *matter domination*. During *recombination*, electrons and nuclei where bound to neutral atoms. Photons were no longer in thermal equilibrium with matter and thus ”decouple”, forming the cosmic microwave background (CMB). Measurements of the CMB today provide one of the most crucial constraints on cosmological parameters.

After recombinations, most photons propagate freely and are quickly redshifted, such that the universe becomes dark. This period is therefore called the *dark ages*. Matter has not yet coalesced into stars and galaxies. The main source of photons visible phtons? is the 21 cm emission line of neutral hydrogen.

Once first stars and galaxies have formed, they emit photons energetic enough to ionize the surrounding hydrogen. This period is called the *epoch of reionization* (EoR). By the end of the EoR, most of the IGM is ionized and continues to be so today.

Some important cosmological params with definitions and their values nowadays, cite PDG

Table 1: Parameters of ΛCDM with estimated values

Name	Symbole	Value	Unit	Source
Hubble constant	H_0	67.4	$\mathrm{km\,s^{-1}\,Mpc^{-1}}$	Planck 2018
Matter density	Ω_m	0.315 ± 0.007		Planck 2018
Dark energy density	Ω_Λ	0.685		Planck 2018
Spectral index	n_s	0.965		Planck 2018
Matter fluctuation amplitude	σ_8	0.811 ± 0.006		Planck 2018
Equation of state	w	-1		Planck 2018

Cite a few papers that give the current status of cosmology in ΛCDM, e.g. Planck 2018, etc.

Cite current values for all params that will be inferred

Need small intro on warm dark matter

How other cosmological models can be tested with the EorFlow setup, maybe also put this somewhere else.

What is matter distribution at the redshift when beginning study?

Small overview of existing studies of timeline of universe, why there is a gap in EoR

Maybe cite more recent paper that explains why studying this exact regime of redshifts

Theoremhe 21 cm Signal

The two possible electron spin states of neutral hydrogen in the 1s state have slightly different energy levels. The transition is forbidden, the excited state having a mean lifetime of $\sim 11 \cdot 10^6$ yr. The parallel state carries slightly more energy, the offset being approximately $\Delta E \approx 10^{-6}$ eV, which corresponds to $\nu \approx 1.42$ GHz, or $\lambda = 21$ cm.

In the following, the singlet state is denoted with index 0 and the triplet state with index 1. Then the index 01 denotes excitation, and 10 de-excitation.

Maybe some more details on ISM at these redshifts?

The 21 cm signal at redshift ranges $z \sim [6 - 30]$ corresponds to a frequency range of $\nu \sim [46 - 200]$ MHz.

This range is covered by the SKA-Low setup, which is sensitive in the $\nu \sim (50 - 350)$ MHz range. https://www.skao.int/en/science-users/118/ska-telescope-specifications

This range belongs to the regieme of radio astronomy. Various background signals disturb the measurement of the 21 cm line in this frequency range:

- The CMB spectrum
- Astrophysical sources
- Foreground noise

Modelling noice of astrophysical sources and of the foreground is in practice the greatest challenge, but the correction for the CMB spectrum can be given in the following.

The 21 cm signal is given in terms of brightness offset temperature δT_B which is calculated from the specific intensity measured by a radio telescope, assuming the Rayleigh-Jeans regieme. Let ν be the frequency at which the 21 cm signal is to be observed. The signal is then composed of the background CMB radiation and the radiation emitted from clouds of neutral hydrogen, minus the CMB radiation absorbed in the neutral hydrogen cloud. This is expressed as

$$\frac{dI}{ds} = -\alpha(\nu) I_\gamma + \alpha(\nu) I_S \tag{10}$$

where I_γ denotes the CMB specific intensity, I_S that of the emission of the 21 cm signal from the neutral hydrogen cloud and $\alpha(\nu)$ the absorption coefficient of neutral hydrogen.

The observered specific intensity I_B is obtained by solving the radiative transfer equation from 10, yielding:

$$I_B = I_S(1 - \exp^{-\tau}) + I_\gamma \exp^{-\tau} \tag{11}$$

$$\tau(\nu) = \int \alpha(\nu) ds \tag{12}$$

where τ is the so-called optical depth. This carries the interpretation that for an optically thick material, $\tau >> 1$, the signal is dominated by the 21 cm signal and for an optically thin material, $\tau << 1$, by CMB radiation.

In the Rayleigh-Jeans regieme, this observed specific intensity is proportional to the temperature. Adding the redshift correction for the CMB temperature, this yields

$$T_{B0}(1+z) = T_S(1 - \exp^{-\tau}) + T_0(1+z)\exp^{-\tau}\,. \tag{13}$$

Finally, the brightness offset temperature is obtained by subtracting the CMB background:

$$\delta T_B = T_{B0} - T_0 = \frac{T_S - T_\gamma}{1+z}(1 - \exp^{-\tau}). \tag{14}$$

The quantities T_S and τ have to be estimated. In practice, this is done numerically. In the following are some comments on the procedure.

The relative number densities of neutral hydrogen are given by Boltzmann statistics:

$$\frac{n_1}{n_0} = \frac{g_1}{g_0}\exp\left(-\frac{\Delta E_{10}}{k_B T_S}\right) \equiv \frac{g_1}{g_0}\exp\left(-\frac{T_\star}{T_S}\right), \tag{15}$$

an expression which defines the excitation temperature, in this context also called spin-temperature T_S. Define spin temp in words.... In thermal equilibrium, this is equal to the thermodynamic temperature of the system.

Physical processes which lead to (de-)excitation of the neutral hydrogen spin state and therefore to a change of T_S are:

1. Interactions with CMB photons
2. Collisions with other particles
3. Wouthuysen-Field effect

More on these interacions? Maybe it's also too much...

The rates for each of these processes are described by the Einstein coefficients, which will be denoted A for spontaneous emission, B for emission/absorption from interaction with CMB, and two effective coefficients C for collision interactions and P for Lyman-α interactions. These measure the rate of interactions in units of probability per unit time per unit spectral energy density.

About thermal equilibrium: quote from 21 cm text: Note that the relevant timescales are all much shorter than the expansion time, so equilibrium is an excellent approximation.

Equilibrium and Rayleigh-Jeans approximation yield:

While coefficients A and B are given by quantum mechanics, the other two...

$$T_S^{-1} = \frac{T_\gamma^{-1} + x_c T_K^{-1} + x_\alpha T_C^{-1}}{1 + x_c + x_\alpha} \tag{16}$$

$$x_c = \frac{C_{10}}{A_{10}} \frac{h\nu}{k_B T_\gamma} \tag{17}$$

$$x_\alpha = \frac{P_{10}}{A_{10}} \frac{h\nu}{k_B T_\gamma} \tag{18}$$

Now, τ has to be estimated. The absorption coefficient measures the probability of an interaction per frequency and unit distance. Given an Einstein coefficient X, it is given by

$$\alpha(\nu) = \frac{h\nu}{c} \; X \; n \; \phi(\nu) \tag{19}$$

where n is number density of atoms partaking in the interaction and ϕ is the line profile normalized to one. Then one finds for neutral hydrogen:

$$\alpha(\nu) = \frac{h\nu}{c}(B_{01}n_0 - B_{10}n_1)\phi(\nu) \tag{20}$$

$$\approx \frac{h\nu}{c} n_0 B_{01} \frac{h\nu}{k_B T_S} \phi(\nu), \tag{21}$$

the approximation coming from equation 15. There are multiple physical effects to be considered in correcting this approximation, and in the end one obtains

$$\tau(\nu) = A(\nu) \frac{x_{HI} n_H}{T_S H(z)} \tag{22}$$

$$A(\nu) = \frac{3c^2 h A_{10}}{32\pi\nu^2 k_B} \tag{23}$$

Using separate estimations for n_H and $H(z)$, one obtains a more detailed approximation of

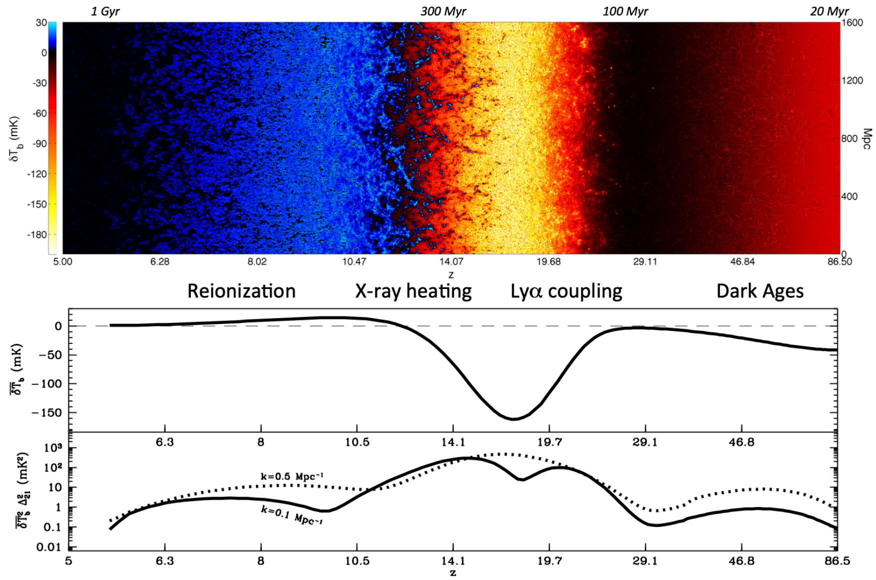


Figure 1: A map of the 21 cm signal of neutral hydrogen

the brightness offset temperature from 14 in the limit $\tau << 1$:

$$\delta T_B \approx 27\, x_{\mathrm{HI}}(1+\delta_b)\left(1-\frac{T_\gamma}{T_S}\right)\left(\frac{1+z}{10}\frac{0.15}{\Omega_{m,0}h^2}\right)^{1/2}\left(\frac{\Omega_{b,0}h^2}{0.023}\right)\,\mathrm{mK} \tag{24}$$

with δ_b the fluctuation of baryon content and x_{HI} the neutral hydrogen fraction.

From the expression in equation 24, it becomes clear that if $T_S > T_\gamma$, the measured brightness offset is positive and vice versa.

Will still have some comments on the figure

1.1.1 Simulating 21cm data

Will explain remaining physical quantities being inferred here

21 cm fast

The parameters used, and the priors on them. Why these parameters? What do they mean physically?

The six parameters and their priors chosen for the simulations in this project are summarized in table 2. The priors are chosen to be uniform and constrained to a range motivated by multimessenger observations.

interpretation of data

Noise simulations

How reioniazation history is calculated

2 Machine Learning Background

There are multiple difficulties in interpreting the 21 cm signal. Traditional Gaussian summary statistics do not capture enough of the signal's structure, reflecting the fact that the processes being observed are highly nonlinear. This also entails that the likelihood of the 21 cm signal is intractible. Additionally, the foreground contamination for ground-based observatories such as the SKA will be significant.

These challenges of non-Gaussianity and foreground noise motivate the use of neural networks for the analysis of SKA data. Another sentence on NNs here. Neural networks are nowadays commonly used in Bayesian inference, which attempts to reconcile prior knowledge and empirical data in order to find optimal parameters describing, for example, our universe.

The issue of an intractible likelihood is tackled with so-called simulation-based inference. The idea behind SBI is to train a NN on simulated data, were the inference parameters are known. Once the model is trained, inference is amortized: the model can perform inference on real data with the pre-trained model inexpensively. The prerequisite is, of course, that the model generalizes well from the training data to the real data. This approach has successfully bin implemented in ergänzen.

Not all models are equally good for a given problem and as a consequence, the choice of model introduces bias to the inference. In an attempt to find a principled way to choose a model, one can use methods developed in information theory, which can here be used to quantify the quality of a model.

In this thesis, a model typically used in parameter inference for simulated SKA-data will be analyzed with the methods of information theory.

Modern physics experiments generate an extreme amount of data, and the SKAO will be no exception. While processing the data will will be difficult because of it's sheer size, an additional obstacle is the complexity inherent in the 21 cm signal, which is highly non-Gaussian. This motivates the use of machine learning to extract physics information from the data. Since there are many possible models which could be applied to the data, there is need for a metric that evaluate's a model's quality.

As an introduction to the methods used in this field, this section covers the concept of neural posterior approximation, some theory behind the class of models analyzed in this thesis as well as a summary of metric from information theory, and how they can evaluate a model's quality.

Traditionally, data as complex as those expected from SKA are analyzed using physics-inspired summary statistics, which map the data to a lower-dimensional space with known interpretation. These will be denoted as $s_{\text{phys}}(\mathrm{D})$. An example is the power spectrum. Such a summary comes with the benefit of interpretability, but the caveat is a potentially large information loss. This motivates the use of a neural networks to learn a different summary $s_\theta(\mathrm{D})$, which is hopefully more informative, though less interpretable:

$$\mathbb{I}(\mathrm{L};\mathrm{D}) \geq \mathbb{I}(\mathrm{L}; s_\theta(\mathrm{D})) > \mathbb{I}(\mathrm{L}; s_{\text{phys}}(\mathrm{D})) \tag{25}$$

In a Bayesian inference task, processing these data would typically be done by choosing:

1. a likelihood $p(\mathrm{D}|\mathrm{P})$ describing how observed data arised from a specific set of parameters,
2. a prior $p(\mathrm{P})$ describing beliefs about the parameter distribution before seeing data.

Given data d, one then calculates the posterior distribution:

$$p(p|d) \sim p(d|p)p(p). \tag{26}$$

In many applied cases, an analytic shape of the likelihood is not available. An example is when the data come from a complex simulator. Then one can sample from this simulator and as such from the likelihood, but the expression $p(d_{\text{obs}}|\theta)$ is not obtainable and as such, the posterior cannot be calculated. In such setting, an alternative is to model the posterior directly. This is known as *likelihood-free inference*. When the posterior is approximated using a neural network, one also speaks of *neural posterior approximation*.

The disadvantage of this method is that one looses...

This doesnt work as well out-of-dataset

An advantage is that unlike with the traditional Bayesian methods, the posterior does not have to be re-calculated given new data: the neural network simply has to be evaluated on the new dataset. In this way, the cost of obtaining the posterior was payed beforehand during training. This concept is known as *amortized inference*. A downside is the fact that one needs data to train the posterior, which is often not available.

As a solution, one can simulate data and train a posterior on this simulated data. This is known as *simulation-based inference* (SBI). Once trained, the posterior can be evaluated on the observed data. The downside is that the choice of the model as well as the choice of the simulator introduce bias to the final estimation and it is unclear wether the model generalizes to real-world data.

2.1 An architecture for complex data

An architecture that has proven to be successful in recent SBI approaches some citations here combines two neural networks in the following principle:

1. The image data are used as input for a convolutional neural network (CNN), which compresses these into a much lower-dimensional summary.
2. This summary is mapped to whichever feature is being inferred by a conditional normalizing flow (CNF).

The idea is that the CNN learns to efficiently encode any relevant physics information, while the flow learns to interpret the CNN's output and map it to the inferred quantity. These two networks are introduced in the following.

Convolutional Neural Networks

CNNs are inspired by the task of implementing neural networks in image analysis, i.e. on data with 2D spatial structure. An early version was successfully implemented in 1983 with Fukushima's "Neucognitron" Neocognitron citation. An architecture implementing backpropagation was LeCun's "LeNet" LeNet citation.

To explain the novel concepts behind the CNN in comparison with a MLP, the CNN architecture will first be discussed in 1D. The generalization to higher dimensions will be discussed afterwards.

Let X denote the data space and Y the feature space.

Let the input have dimensions $x \in \mathbb{R}^{C\times H\times W}$ where C the channel dimension, H the height dimension and W the width dimension. In case of an RGB image, this channel dimension would be three.

The CNN learns a function $f_\theta : X \to Y$.

As opposed to a simple MLP, the CNN offers the following advantages:

Using summation convention

Not writing bias explicitly

A basic MLP consists of only one type of layer. If h is the output of the previous layer, the current layer is defined by

$$z_i = \phi(W_{ij}h_j), \tag{27}$$

The CNN implements additional layers.

A convolutional layer applies a convolution operaton:

Talk abt convolution vs cross-correlation here

Is this def even correct? Not need different indices on h?

$$z_i = \phi([w \circledast h]_i) = \phi(\sum_k w_k h_{i+k}) \tag{28}$$

where the dimensionality of the kernel $w \in \mathbb{R}^K$ is typically much lower than that of the data.

A pooling layer applies a pooling operation:

$$z_i = \operatorname{pool} K_{ij}h_j \tag{29}$$

where K defines a pooling window for each index i. Typical choices for pool are maxpool, which chooses the maximum of all given values, or meanpool, which calculates their mean.

The purpose of a pooling layer is to...

Add normalization layers here

Add info on stride params etc here

Hier kann man schon ein bisschen mehr schreiben, this is all very basic. More info on the CNN specific to this project to come.

Put it together here, potentially add image of total architecture.

Also missing MSE loss function as usual loss

Conditional Normalizing Flows

A normalizing flow is a model that learns to transform a simple (often Gaussian) distribution into a more complex one, the target distribution. The transformation is chosen to be invertible, such that the model becomes generative: once trained, it can be inverted and by transforming samples from the normal, one can sample from the approximated target distribution.

Let $u \sim p_u = \mathcal{N}(0, I_n)$ with $n \in \mathbb{N}$ the base distribution and $x \sim p_x$ with p_x the n-dim. target distribution. Let $f : \mathbb{R}^n \to \mathbb{R}^n$ a diffeomorphism, with $g = f^{-1}$ its inverse. The training objective is to find g, such that

$$u = g(x) \sim p_u, \qquad x \text{ sampled from } p_x. \tag{30}$$

The process of generating samples from the approximated distribution is the inverse of the above:

$$x = f(u) \sim q_x, \qquad u \text{ sampled from } p_u. \tag{31}$$

For a given transformation, one computes $p_x(x)$ by applying the change of variables formula:

$$q_x(x) = p_u(g(x)) \left| \det \left(\frac{\partial g(x)}{\partial x} \right) \right| = p_u(u) \left| \det \left(\frac{\partial f(u)}{\partial u} \right) \right|^{-1}. \tag{32}$$

The NLL is the loss function used for optimization, which then becomes:

$$\mathcal{L}_{\text{NF}} = -\mathbb{E}_{x \sim p_x} \left[\frac{|g(x)|^2}{2} + \log \left| \det \left(\frac{\partial g(x)}{\partial x} \right) \right| \right] \tag{33}$$

where the first term is a regularization for the latent (in this case Gaussian) space and the second stems from the transformation. See cite Plehns book for more details.

The normalizing flow becomes expressive by allowing the transformation to be as complex as possible, while restricting it to a class of functions with efficiently computable Jacobian determinants makes it computationally tractable. In practice, one often models g as a composition of simple neural networks, e.g. multilayer perceptrons. How these are invertible...

Next few lines could also just be skipped

Let $g_\theta = g_1 \circ g_2 \circ ... \circ g_k$ with $g_i : \mathbb{R}^n \to \mathbb{R}^n, x \mapsto g_i(x|\theta_i)$ and $k \in \mathbb{N}$.

Now the loss becomes

$$\text{NLL}_\theta = \sum_{i=1}^{k} \log |\det J(g_i)| - \log p_u(g_\theta(x)) \tag{34}$$

How to ensure it is a probabilistic model?

Come up with nicer notation for $g_i(\cdot|\theta_i)$

Now what this specific Freia flow does. Research on website!

Or maybe put exact architecture in appendix?

What about the B-INN in Plehns book?

2.2 Information Theory and Neural Estimators

The model discussed in the previous section can be implemented with various different architectural choices. One important choice is that of the *bottleneck size*, that is, the dimensionality of the CNN's output. This part of the model is referred to as the bottleneck because it is the point in which the data are compressed to the highest degree. While a larger bottleneck should be able to capture more information about the data, one needs a principled way to balance the increased model complexity with the increase in information captured by the model. One way to do this is to evaluate the *mutual information* between the feature and the bottleneck space across different such bottleneck dimensions. Mutual information and it's estimation are introduced in the following sections.

Some Basic Measures in Information Theory

Given a random variable $X \sim p$, the *entropy* of X is defined as:

$$\mathbb{H}(X) = -\mathbb{E}_X\left[\log p(X)\right] \geq 0. \tag{35}$$

If the logarithm is chosen as with base 2, one defines the unit of this quantity as bits. If one uses the natural logarithm, the entropy is measured in units of nats.

Entropy is a measure of the uncertainty of a random variable. A high entropy corresponds to a lack of predictability of the random variable's outcome. As an example, the entropy of the uniform distribution is maximal, while that of the delta distribution is zero. One therefore considers entropy to be a measure of the amount of information contained in the random variable's distribution.

The generalization to conditional entropy is:

$$\mathbb{H}(X|Y) = -\mathbb{E}_{X,Y}\left[\log p(X|Y)\right] \tag{36}$$

Another important metric in information theory is the *Kullback-Leibler divergence*. Let $X \sim p$ and $Y \sim q$ be two random variables over the same sample space. The KL-divergence between

p and q is defined as:

$$\mathbb{D}_{\mathrm{KL}}(p||q) = \mathbb{E}_X \left[\log p(X) - \log q(X)\right] \geq 0 \tag{37}$$

The KL-divergence satisfies the identity axiom, $\mathbb{D}_{\mathrm{KL}}(p||q) = 0 \Leftrightarrow p = q$. It is not symmetric and does not satisfy the triangle inequality and as such is not a metric. However, it does still carry an interpretation as a distance between two probability distributions: from the above expression it is clear that the more q resembles p, the smaller the measured KL-divergence becomes. A smaller the KL-divergence between the two distributions means there is less information loss by using q to approximate p.

Maybe a word about consequece of assymetry of D KL

Minimizing the forwards KL-divergence between the data distribution and the model distribution is equivalent to maximizing the negative log-likelihood, because

$$\mathbb{D}_{\mathrm{KL}}(p_{\mathrm{data}}||q_\theta) = -\mathbb{H}(p_{\mathrm{data}}) - \mathbb{E}_{x\sim p_{\mathrm{data}}}\left[\log q_\theta\right] = \mathrm{NLL} + \mathrm{const} \tag{38}$$

Lastly, the *mutual information* between two random variables X and Y is defined as:

$$\mathbb{I}(X;Y) = \mathbb{D}_{\mathrm{KL}}(p(X,Y)||p_X(X)p_Y(Y)) \geq 0. \tag{39}$$

An alternative formulation in terms of entropy is:

$$\mathbb{I}(X;Y) = \mathbb{H}(X) - \mathbb{H}(X|Y) = \mathbb{H}(Y) - \mathbb{H}(Y|X). \tag{40}$$

Mutual information is symmetric and nonnegative. It is zero iff X and Y are independent and becomes maximal iff $X = Y$. One interprets it's value as the amount of information obtained about one random variable while observing another.

Something about how MI is sensitive to higher moments, maybe this can tease why one needs X amount of samples in MI estimation?

Mutual Information Estimation

In the context of representation learning, mutual information can be used as a measure of model quality. This is motivated by the *data processing inequality*, which in this context can be phrased as follows:

Let X, Y and Z be random variables where X is the original data, Y is a learned summary of X and Z is a label associated with X. These form a Markov chain $X \to Y \to Z$, where Z is only inferred from Y. Then Z cannot contain more information about X than Y, so

$$\mathbb{I}(X;Y) \geq \mathbb{I}(X;Z) \tag{41}$$

A good representation keeps as much information about the original data as possible, subject to the constraint that it should contain less total information than the data itself, i.e. summarize it in some way. Mit Formel unterstützen

It also captures more information than just covariance!

Something on summary statistics?

In practice, this necessitates an estimation of mutual information between the representation and the target distribution.

In the setting of variational inference, one can use the learned posterior to estimate mutual information in a natural way, described in the following:

$$\mathbb{I}(X;Y) = \int_{\mathcal{X}\times\mathcal{Y}} p(x,y) \log \frac{p(x|y)}{p(x)} \ , \tag{42}$$

where the distribution $p(x|y)$ is estimated by a learned $q_\theta(x|y)$, the optimization objective being to minimize the KL-divergence $\mathbb{D}_{\mathrm{KL}}(p||q_\theta)$. The Monte-Carlo approximation of equation (42) using this distribution is referrred to as the *Barber-Agakov estimate*

$$\hat{\mathbb{I}}(X;Y) = \mathbb{I}_{\mathrm{BA}}(X;Y) = \mathbb{E}_{P(X,Y)}\left[\log q_\theta(x|y) - \log p(x)\right] \tag{43}$$

This is computable, because the distribution of the prior $p(x)$ can in practice either be ignored as a constant factor, or in case of SBI is chosen and therefore known. If this is not the case, $p(x)$ can also be estimated separately, which introduces more uncertainty in the above estimate.

Because the KL-divergence is non-zero, the BA-estimate yields a lower bound on the true mutual information:

$$\mathbb{I}(X;Y) \geq \mathbb{I}_{\mathrm{BA}}(X;Y) \tag{44}$$

Estimating the error of this estimate from the true value?

Maybe something about the bias of the estimator oder so...

Why estimating MI in high dimensions is hard, and quantify!

Also another method of approximation, maybe Donsker-Varadahn (MINE) ?

Maybe something on limitations on the interpretation of these metrics. Could also tease limitations in high dimensions, or give intuition for amount of data needed.

3 Implementation

Training is performed on one NVIDIA A30 GPU. Need specific values of learning rate etc for both models. Should this go in appendix?

3.1 The dataset

Simulating the data used for inference is not part of this project. An existing dataset is used. The data were simulated with 21cmFast, cite the paper somewhere!. The parameters varied in the simulation are:

- The matter density Ω_m, describing the fraction of the universe's energy density from matter (incl. dark matter)
- The warm dark matter mass ω_{WDM}, describing the mass of a hypothetical warm dark matter particle
- The minimum virial temperature T_{vir}, describing the minimum virial temperature to enable star formation in the first places
- The ionization efficiency ζ, characterizing the strength of ionizing radiation
- The specific X-ray luminosity L_X, a measure of total X-ray luminosity of galaxies, measured per star formation rate for comparability
- The X-ray energy threshold for self-absorption by host galaxies E_0, a measure for the energy of X-rays below which these do not escape their host galaxies.

Could still comment on the significance of each parameter. Why are X-rays so relevant?

Do I need log priors or not?

Their priors used on each of these parameters are shown in table 2. Add 2018 Planck estimates here instead?. All other parameters of the flat ΛCDM cosmology are fixed to the Planck measurements. This means $\Omega_b = 0.04897$, $\sigma_8 = 0.8102$, $h = 0.6766$, and $n_s = 0.9665$.

The final dataset consists of 5000 sets of lightcones with the corresponding simulation parameters and the neutral hydrogen fraction. Each lightcone is originally $140 \times 140 \times 2350$ voxels in size, corresponding to a redshift range of $z \sim [5, 12]$ find out spacial dimensions.

The neutral hydrogen fraction is calculated from the lightcone and stored as a 92-dimensional vector. The dataset is split into a training, validation and test set with relative sizes 0.7, 0.2 and 0.1, respectively.

See comments on redshift range in Benedicts paper

Table 2: Parameters used for the simulations

Name	Symbol	Prior
Matter density	Ω_m	$\mathcal{U}(0.2, 0.4)$
Warm dark matter mass	ω_{WDM}	$\mathcal{U}(0.3, 10)$ keV
Minimum virial temperature	T_{vir}	$\mathcal{U}(10^4, 10^{5.3})$ K
Ionization efficiency	ζ	$\mathcal{U}(10, 250)$
Specific X-ray luminosity	L_X	$\mathcal{U}(10^{38}, 10^{42})$ erg/s/$\mathrm{M}_\odot \times$ yr
X-ray energy threshold for self-absorption by host galaxies	E_0	$\mathcal{U}(100, 1500)$ eV

3.2 The network architecture

Insert sentence introducing this section. Also, find better section title

A CNN learns a compression of lightcones to a much lower-dimensional latent space. A normalizing flow learns a mapping conditioned on this latent space to the target distribution.

Something about representation learning?

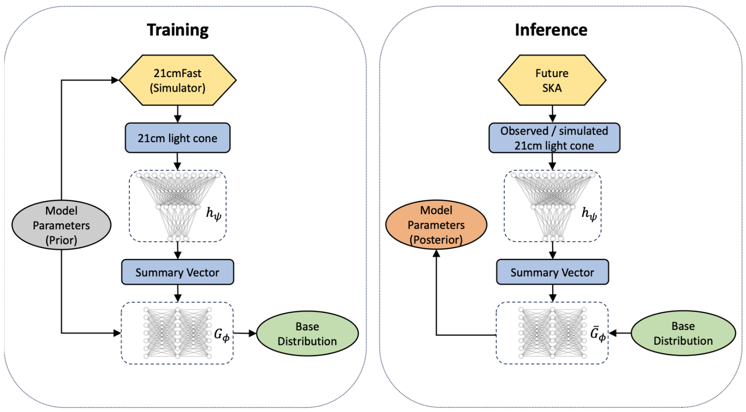
Examples for target distributions include the parameters used for the simulations, or a simple physical summary statistic of the image data.

Which construction is PIE-INN based on? Is it completely new?

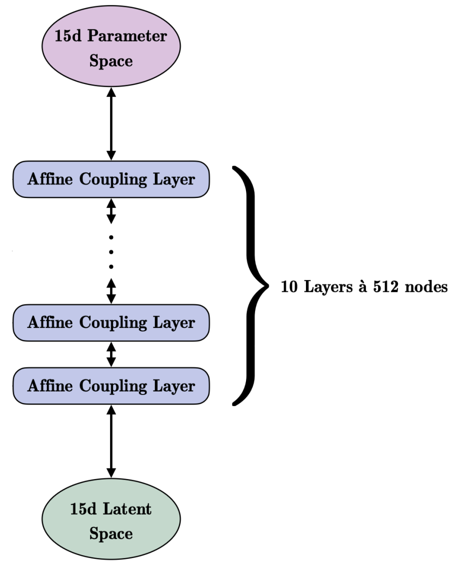
One such construction is *21cmPIE-INN*, which consists of a CNN acting as a compressing network and a normalizing flow mapping its output to six parameters used for the original simulations. The architecture is visualized in figure 2a.

It is also possible to learn e.g. the reionization history with variable resolution from the CNN's summary, as is done with the *EoRFlow* architecture Citation here. The flow implemented in EoRFlow is slightly more complex the the 21cmPIE-INN. It's architecture is visualized in figure 2b.

Let x denote an image from lightcone distribution, θ the corresponding element of the label distribution, s_ψ the CNN and g_ϕ the transformation defining the normalizing flow. The network's parameter indices are surpressed in the following. For both inference cases, the



(a) Architecture of 21cmPIE-INN, taken from citation



(b) Architecture of the flow used in EoRFlow, taken from citation

Figure 2: Overall caption for both figures

loss function for jointly training the two models is

$$\mathcal{L}_{\text{NPE}} = -\mathbb{E}_{(x,\theta)\sim p(x,\theta)}\left[\frac{|g(\theta|s(x))|^2}{2} + \log\left|\det\left(\frac{\partial g(\theta|s(x))}{\partial\theta}\right)\right|\right]. \tag{45}$$

Some introductory sentence...

Table 3 shows the network architectures of the CNN and the NF for both inference cases Shape values still taken from Caroline's paper, check these still! In the CNN, the first convolutional layer is designed to learn patterns in the redshift direction. An aggressive stride is used to alter the dimensionality of the next layers. The kernel for hidden layers is set to (3x3x2) and max pooling is only applied in spacial dimensions. Finally, all information is compressed into a 128-dimensonal feature vector, with global average pooling, which is in turn transformed into an N_B-dimensional output vector with a simple MLP. In case the CNN is pretrained, an additional MLP head can be attached, which maps the N_B-dimensional output to a 6-dimensional output. The head can be removed for the rest of training. This has the benefit of leaving the rest of the architecture untouched. The CNF gets the CNNs N_B-dimensional output as conditional input and learns to transform the feature space to a gaussian of the same dimension. It is used in two variants depending on the inference task, the heavier of the two architectures containing more coupling layers and more nodes per layer.

Table 3: CNN network architecture

Layer	Shape
Input Layer	(140,140,1916,1)
3x3x75 Conv3D	(138,138,23,32)
3x3x2 Conv3D	(136,136,22,32)
2x2x1 Max Pooling	(68,68,22,32)
3x3x2 Conv3D	(66,66,21,64)
1x1x0 Zero Padding	(68,68,21,64)
3x3x2 Conv3D	(66,66,20,64)
2x2x1 Max Pooling	(33,33,20,64)
3x3x2 Conv3D	(31,31,19,128)
1x1x0 Zero Padding	(33,33,19,128)
3x3x2 Conv3D	(31,31,18,128)
Global Average Pooling	(128)
Dense	(128)
Dense	(128)
Dense	(128)
Dense	(N_B)
(Dense)	(128)
(Dense)	(128)
(Dense)	(6)

Number of Parameters: 651,526

Table 4: Flow network architecture

Light network variant

Feature	Shape
Input layer	N_B
Coupling layers	8
Nodes	256
Subnet depth	3

Number of Parameters: 651,526

Heavy network variant

Feature	Shape
Input layer	N_B
Coupling layers	10
Nodes	512
Subnet depth	3

Number of Parameters: 651,526

3.3 Inference of the neutral hydrogen fraction

For inference of the neutral hydrogen fraction, the lightcones are preprocessed as in reference yannics paper: The temperature fluctuation values are min-max-normalized individually for each lightcone Motivation from CNN? and the lightcone is truncated to $140 \times 140 \times 1916$ voxels. This allows the labels to first be truncated to 75 dimensions and then binned to 15-dimensional vectors. This step reduces the dimensionality of the inference problem. Any additional reason for lightcone truncation? An additional logit transformation is applied to the labels, which improves training performance of the flow. A sigmoid transform can be applied during inference to map the labels back to their physical state. The total dataset has a size of insert current size.

In order to model the conditional distribution $p(x|y)$, an ensemble of models is trained for each bottleneck dimension. Each ensemble consists of multiple training runs at different weight initializations, i.e. different seeds. The CNN and flow are trained jointly. Both models are trained using an optimizer, a scheduler and a joint scaler. The optimization method is AdamW, the scheduler method is ReduceLROnPlateau and torch's GradScaler ensures more stable backpropagation. A summary of all the hyperparameters and their chosen values can be found in Cite Appendix. Maybe talk about Optuna?.

Talk about significant difficulties with training here, and data methods used to improve training speed. Also mention gradient accumulation.

Unlike the original distribution of simulation parameters, the distribution of reionization history vectors is not uniform. It therefore has to be modeled separately. It is modeled with a flow of the same architecture as the one from the conditional distribution, without any conditional input. An additional prior model adds uncertainty to the mutual information estimation. The idea behind choosing the same architecture to model the label distribution is that an artefact of the flow $q_\theta(x|y)$ leading to high variance in the MI estimation might be learned by $q_{\tilde{\theta}}(x)$ as well, thus leading to smaller variance in the final MI estimation.

3.4 Inference of the simulation parameters

For inference of the simulation parameters, the data are preprocessed as in reference benedicts paper: The lightcones are kept in their original dimensionality and lightcones as well as labels are globally standardized.

Talk about adjusting CNN architecture here

The conditional distribution $p(x|y)$ is again modelled with an ensemble of models for different bottleneck dimensions. Training is done with AdamW optimization and the ReduceLROnPlateau scheduler. No scaler is needed. Using gradient accumulation with an effective batch size of 32 causes training to fail, therefore gradient accumulation is not implemented. The model does not converge if the CNN and flow are trained jointly from the beginning, so a three-step training process is implemented. First, the CNN is pretrained on the labels with an MSE loss. Since this has to be done for varying bottleneck dimensions, an additional MLP head is added to the existing CNN architecture, mapping the N_{dim} output to 6 dimensions. Pretraining ends after 50 epochs. The CNN weights are then frozen and the MLP head is removed for the rest of training. The CNN outputs after pretraining are then used to pretrain the flow for 120 epochs. Afterwards, the CNN weights are unfrozen and the optimizers and schedulers of both models are reset. Both models are then trained jointly for another 80 epochs.

4 Results

Still gathering some more results, but MI estimations won't change much anymore probably.

Might be nice to evaluate some metric for inference performance that Yannic used as well, to show that I recovered the results.

A word on bootstrap error calculations for comparison

Each model is trained for 250 epochs, with an average runtime of insert average runtime. Due to these training times, only specific bottleneck dimensions are analyzed. Along with varying bottleneck dimensions, a number of different weight initializations (seeds) are tested. The held-out test set consisting of 10% of the total dataset (500 samples) is used to obtain estimates on the conditional entropy as well as the mutual information:

$$\hat{\mathbb{H}}_C = -\mathbb{E}_{p(l,s)}\big[\log q_\theta(l|s)\big] \tag{46}$$

$$\hat{\mathbb{I}} = -\mathbb{E}_{p(l,s)}\big[\log q_\theta(l|s) - q_\phi(l)\big] \tag{47}$$

For each training run, the CE/MI estimation is done on the model with the best validation performance.

Two sources of error in the estimation are considered. The error of the individual model, stemming from the variability of log probability values on the test set, is referred to as the *model error*. This is estimated from the standard deviation and confirmed by bootstrap analysis, see below. When comparing results across seeds, an additional *seed error* must be considered, which is also estimated from the standard deviation of the results.

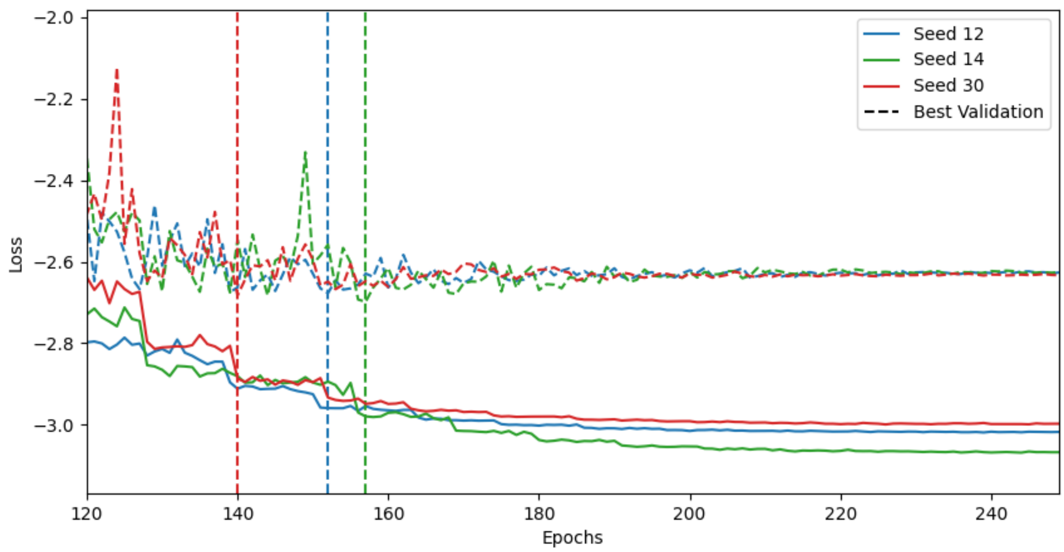


Figure 3: Train and Validation Losses for Bottleneck Dimension 15

As an example, the training and validation losses for an ensemble of runs with bottleneck dimension $N_{\text{dim}} = 15$ are shown in figure 3. Vertical lines indicate the epoch with best validation performance. Observations... The loss plots for all other ensembles can be found in the appendix Reference coming soon. In order to compare the training processes, table 5 summarizes losses, epochs until convergence and seed variance for different bottleneck dimensions. One observes a higher seed variance for models with smaller bottleneck dimensions. There is also a higher variability in their training and validation losses across seeds. This indicates a higher training instability for models with smaller bottleneck dimension.(Table almost ready, results coming soon) As an example, figure **??** showcases how the conditional entropy esti-

Table 5: Results on training stability

Bottleneck	Average best epoch	Average best val loss
4	130	0.39.
6	38.49 ± 0.95	0.83
15	39.90 ± 0.42	0.13
16	39.57 ± 0.46	0.13

mate evolves during model training for the ensemble with bottleneck dimension $N_{\text{dim}} = 15$. Vertical lines again indicate the epoch with best validation performance. All models reach a saturated estimate of the conditional entropy in the epoch range... The conditional entropy curves showcase a higher instability than the validation curves, which is attributed to the

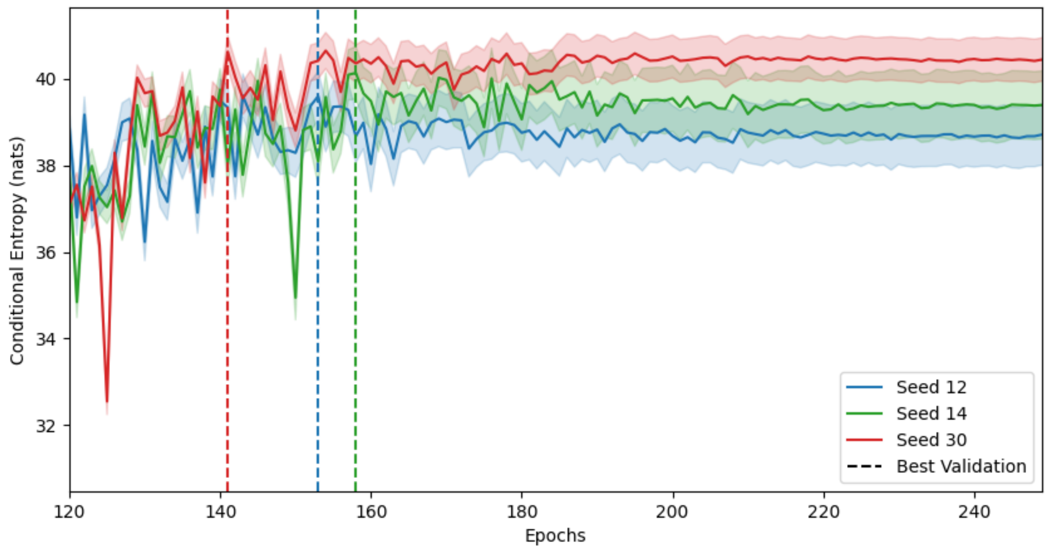


Figure 4: Conditional Entropy per Epoch for Bottleneck Dimension 15

smaller size of the test set. All other ensemble plots are in the appendix (reference coming soon). The model error on the conditional entropy estimate grows with increasing training steps. Some reionization histories are attributed with very low likelihood, such that the distribution of $\log q_\theta(\text{label}_i|\text{summary}_i)$ has a heavy left-sided tail. This is in parts because of the distribution of the labels themselves.

Table 6 shows the estimated conditional entropy for each model. All entropy values are given in nats. observations...

Table 6: CE estimates across bottleneck dims N_{dim} in nats

N_{dim} \ Seed	12	14	30
4	38.88 ± 0.89	-	37.68 ± 0.42
6	36.75 ± 0.40	-	40.25 ± 0.70
15	39.64 ± 0.52	40.15 ± 0.59	40.61 ± 0.37
16	39.81 ± 0.38	40.17 ± 0.50	39.72 ± 0.68
25	-	-	40.22 ± 0.57

A separate normalizing flow is trained on the label distribution. The model with best validation performance is chosen for the following analysis. It's distribution of $\log \tilde{q}_\theta(\text{label}_i)$ also showcases a heavy left-sided tail.

As an example, a histogram plot of the distributions $\log q_\theta(\text{label}_i|\text{summary}_i)$ and $\log \tilde{q}_\theta(\text{label}_i)$

Table 7: MI estimates across bottleneck dims N_{dim} in nats

N_{dim} \ Seed	12	14	30
4	6.00 ± 0.73	-	4.76 ± 0.31
6	3.87 ± 0.30	-	7.24 ± 0.61
15	6.70 ± 0.45	7.25 ± 0.48	7.74 ± 0.25
16	6.94 ± 0.29	7.31 ± 0.39	6.81 ± 0.57
25	-	-	7.34 ± 0.47

is shown in figure insert figure.

The final estimates of conditiona entropy with errors as shown in table 9. The total estimated errors as well as separate seed and model errors are listed.

Table 8: Final Esimation of CE per Bottleneck Dimension in nats

Bottleneck	CE	Seed error	Model error	No. samples
4	38.26±0.49	0.36	0.34	2
6	38.47±1.02	0.99	0.27	2
15	40.10±0.25	0.17	0.19	3
16	39.88±0.26	0.11	0.23	3
25	40.20±0.58	-	0.58	1

Table 9: Final Esimation of MI per Bottleneck Dimension in nats

Bottleneck	MI	Seed error	Model error	No. samples
4	5.39±0.45	0.35	0.28	2
6	5.58±1.01	0.99	0.22	2
15	7.23±0.22	0.16	0.14	3
16	7.02±0.21	0.10	0.19	3
25	7.31±0.49	-	0.49	1

These results can be described as follows: One does observe a general trend of increasing mutual information for increased bottleneck dimension, though the increase is not monotonous. Seed and model error are similar in magnitude, though with exceptions in case of failed training runs.

5 Interpretation

Still gathering some ideas for the interpretation...

Possible interpretations of the results:

The flow seems to be overcondident, because of the long negative tail in log probability plot. If choosing MI estimation from earlier epochs, get smaller error! This is due to observed overfitting and still improving performance.

The 500 samples are simply not enough to be a precice probe of the distributions being compared. On the other hand, could not have taken more from training set because this would have caused even more training instability. Especially the large seed error points towards this. In total, the 5000 samples seem to be borderline to successfully compare two distributions as high-dimensional as these...

Alternative interpretation:

The CNN learns the latent space. If the weights of the CNN aren't updated enough and the flow is doing too much of the heavy lifting, then the latent space structure might be hard to estimate, resulting in long negative tail.

Check to see wether the extreme models are the reason for this, or if this is a problem with the flow.

The seed uncertainty for this model is very high, given some numbers here. The high sensitivity towards weight initialization points towards unstable training.

One possible explanation for this instability could be the choice of hyperparameters. For example, a large learning rate might magnify an initially chaotic training process, causing the model to end in different minimia of the loss landscape.

Another possibility is the choice of architecture. For example, the flow might be too deep given the data. While one might expect the need for a deep architecture to infer 15-dimensional labels, these labels are likely highly correlated across dimensions because of curve shape etc, making the problem easier than naively expected. The fact that one observes an early saturation in mutual information across bottleneck sizes supports the latter statement.

Of course, there is always the possibility of there being too little training data. Choosing larger parts of the training set for validation and testing indeed showed an increase in overfitting towards the end of training.

Also comment on error changed between CE and MI estiamte

Something about consistency of MI estimate, since e.g. CE estimate is not larger zero from beginning. Maybe cite from paper?

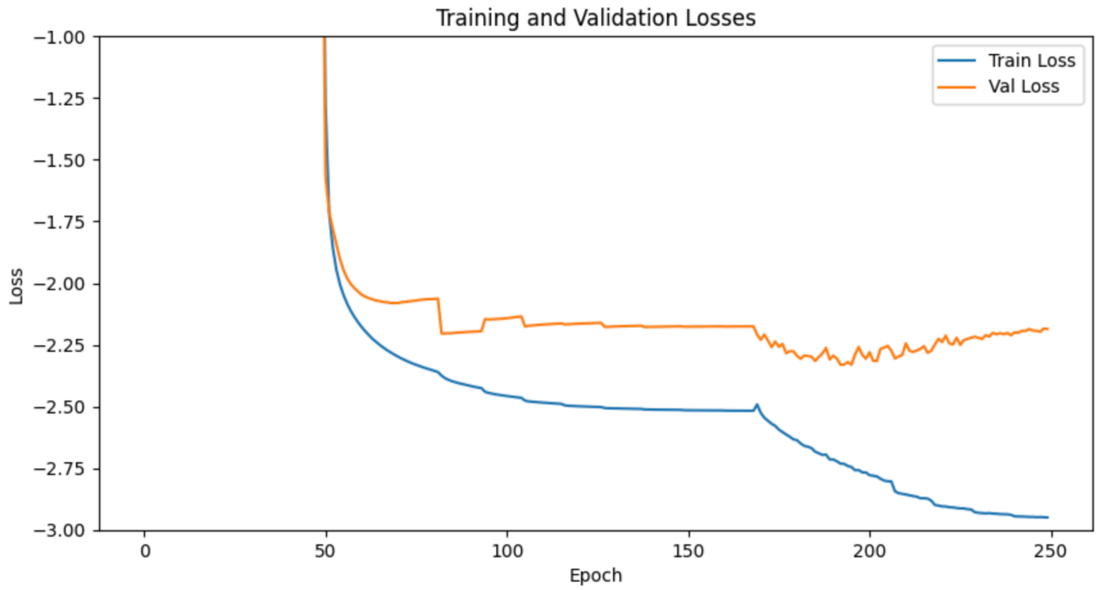


Figure 5: Joint loss for bottleneck dim 5

From paper: ”In par- ticular, higher estimated MI between observations and learned representations do not seem to indi- cate improved predictive performance when the representations are used for downstream supervised learning tasks (Tschannen et al., 2019).”

6 Outlook

Hi Caroline, in case you're already reading this, I finally got the CNN to train on the 6-param inference problem as well! I can now pretrain with a smaller bottlenck too and I have my first successful training runs, I'm still implementing that in the thesis and I'm excited to see the results tonight (Sat.) ◡̈

See 5 for an example of joint training loss after pretraining both models on a bottleneck dimension of 5.

How one could incorporate information geometry for similar problems

Any idea how to numerically resolve F_{ij} for this flow?

Alternative choices than to model instead of $p(x|y)$ and $p(x)$?

The compressing network usually has a much heavier architecture. Once it has been trained, one can learn different lighter networks to map its summary to different target distributions. This is only really true in the self-supervised case, see SKATR. So Maybe this should go in outlook

A Network Implementation

Network architecture, training details, losses